

EC571: Digital VLSI Circuit Design

Homework 1 – Fall 2026

Total: 100 points

Due: September 23 at 11:59 pm ET

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 1 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all intermediate reasoning. A numerical answer, Boolean expression, or circuit diagram without explanation will receive limited credit.

For transistor-level questions, clearly indicate which transistors are ON/OFF for the input cases you analyze. For design questions, state your assumptions and justify your design choices.

You may use AI tools only for concept review. You may not use AI tools to produce final answers, final schematics, or final derivations. At the end of your submission, include one sentence disclosing whether and how you used AI tools.

C_{sw} = total switched capacitance

$$= N \times \cancel{\alpha} \times C$$

$\uparrow$ # of gates $\uparrow$ Activity factor $\uparrow$ Avg. C of single gate.
 ignore $\downarrow$ - yang wu

NOTES PAGE
1 for Q1a

$$P_{dyn} = \alpha (N \cancel{\alpha} C) \times V_{dd}^2 \times f. \quad \leftarrow ?? \text{ asked Piazza's Yang Wu}$$

$$\frac{P'_{dyn}}{P_{dyn}} = \frac{\alpha \times (1.3 N \times 0.65 C) \times (0.8 V_{dd})^2 \times 1.25 f}{\alpha \times (N \times C) \times V_{dd}^2 \times f}$$

$$P_{dyn} \neq \alpha \times (N \times C) \times V_{dd}^2 \times f$$

$$= \frac{\alpha \times 1.3 N C \times 0.64 V_{dd}^2 \times 1.25 f}{\alpha N C \times V_{dd}^2 \times f}$$

$$= 1.3 \times 0.64 \times 1.25$$

$$\frac{P'_{dyn}}{P_{dyn}} = 1.04.$$

P_{dyn}

→ Increased transistor count increases P_{dyn} proportionally as P_{dyn} increases by total switched capacitance increasing increases by number of transistors in a block.

→ Similarly, increasing average capacitor by transistor increases P_{dyn} as C is proportional to C_{sw} is proportional to P_{dyn} .

→ P_{dyn} is proportional to V_{dd}^2 , i.e. it is 'proportional to the square'. This is due to the V_{dd} in the direct power approx. equation.

→ frequency is directly proportional - also in P_{dyn} formula as simple multiplier.

Question 1: From Technology Scaling to Design Constraints [20 points]

This question asks you to reason about why VLSI design is not simply “make transistors smaller.”

A chip design team is comparing two technology generations. In the older generation, a digital block contains N transistors, has average switched capacitance C , operates at supply voltage V_{DD} , and runs at frequency f . In the newer generation, the team expects the following first-order changes:

$$N' = 2N \quad \# \text{ of TRANSISTORS}$$

$$C'_{\text{per transistor}} = 0.65 C_{\text{per transistor}} \quad \text{AVG SWITCHED CAPACITANCE}$$

$$V'_{DD} = 0.8 V_{DD} \quad \text{SUPPLY VOLTAGE}$$

$$f' = 1.25 f \quad \text{FREQUENCY}$$

Assume the activity factor is unchanged. Use the dynamic power approximation:

$$P_{\text{dyn}} = \alpha C_{\text{sw}} V_{DD}^2 f$$

where C_{sw} is the total switched capacitance.

(a) **Dynamic power scaling.** Derive the ratio

$$\frac{P'_{\text{dyn}}}{P_{\text{dyn}}}$$

Clearly show how transistor count, capacitance per transistor, voltage, and frequency each affect the result. [5 points]

See Notes Page 1 for
answers to Q1a :]

Energy per operation = $\frac{\text{dynamic power}}{\text{frequency}}$ ← ops per second

$$E = \frac{\propto NC(V_{dd}^2)f}{f}$$

f cancels out?
~~It might~~ could be misunderstanding the question - Ask Yang Wu.

$$E = \propto NC V_{dd}^2.$$

$$\frac{E'}{E} = \frac{\propto N' C' V'^2_{dd}}{\propto N C V^2_{dd}} = \frac{2N \times 0.65 C \times (0.8)^2 V_{dd}}{N C V^2_{dd}}$$

$$= 2 \times 0.65 \times 0.8^2 = 0.832$$

$$\frac{E'}{E} = 0.832.$$

Q1B,
Note page 7

(1c)

Answer from Pt. 1A: $\frac{P'_{dyn}}{P_{dyn}} = 1.04.$

$$\overset{\substack{\uparrow \\ \text{new } A}}{A'} = 0.7 \overset{\substack{\uparrow \\ \text{old } A}}{A}$$

$$\frac{P'_{dyn} / (0.7A)}{P_{dyn} / A} = \frac{1.04 / 0.7A}{1 / A} = \frac{1.04}{0.7A} \times A = \frac{1.04}{0.7}$$

$$\frac{P'_{dyn} / A'}{P_{dyn} / A} = \frac{1.04}{0.7} = 1.4857 \text{ (4dp)}$$

$\frac{104}{70}$ $\uparrow$ 52/35

~~Notes~~ Notes Page 6
Q1c

- (b) **Energy per operation.** Assume the number of useful operations per second scales proportional to frequency only. Derive the ratio of energy per operation in the newer generation relative to the older generation. [4 points]

$$\frac{E'}{E} = 0.832$$

Working on note page 7

- (c) **Power density.** Suppose the chip area scales by $0.7\times$, meaning

$$A' = 0.7A.$$

Using your result from part (a), derive the ratio of power density:

$$\frac{P'_{\text{dyn}}/A'}{P_{\text{dyn}}/A}$$

See Notes page 6
for working -

[4 points]

$$\frac{P'_{\text{dyn}}/A'}{P_{\text{dyn}}/A} = \frac{1.04/0.7A}{1/A} \Rightarrow \frac{1.04}{0.7} = \frac{52}{35}$$

- ① The dynamic power approximation is higher - this on its own isn't good (unless the tradeoffs are worth it)
- ② The energy per operation has decreased from the old generation to the new - great! more efficient! however, why is the amount of energy used higher?
- ③ The power density has increased significantly, due to the (impressive) decrease in area. This means that the amount of energy (therefore heat) is far higher than the older version, requiring lots of cooling. This is not great.
- ④ Presumably the design is far more complex, to fit twice the number of transistors into 0.7x the area - but the newer technology is certainly not automatically better. As Prof. Joshi says, "it depends!"

Q1d, notes Pg 8

- ④ complex design - entire die image transferred @ once & resolution improving all the time.
& processed

Non Recurring cost $\uparrow$ every technology generation

Constraints:
 Resolution, alignment,
 Density. Parasitics.

Abstraction Device $\rightarrow$ circuit $\rightarrow$ gate $\rightarrow$ module $\rightarrow$ system

reasonable & acceptable amounts of failure $\rightarrow$ space to optimise

$\hookrightarrow$ Reduce frequency for ~~increased~~ reduced power consumption?

different bonding techniques - heat dissipation, cost

$\hookrightarrow$ Tape - automated bonding $\leftarrow$

$\hookrightarrow$ flip chip bonding

surface mount - higher density

Q1E notes
Pg 8

- (d) **Design interpretation.** Based on your answers, explain whether the newer technology is automatically "better." Your answer must discuss at least three of the following: performance, energy, power density, cooling, reliability, and design complexity. [4 points]

Previous answers:

$$\frac{P'_{dyn}}{P_{dyn}} = 1.04$$

$$\frac{E'}{E} = 0.832$$

$$\frac{\text{Power density}}{\text{Power density}} = \frac{52}{35} \approx 1.4857.$$

Please read
Notes Pg 8,
Q1D

- (e) **Course connection.** Identify two topics from the first EC571 lectures that help address the design challenges revealed by this scaling exercise. For each topic, explain in one sentence why it matters. [3 points]

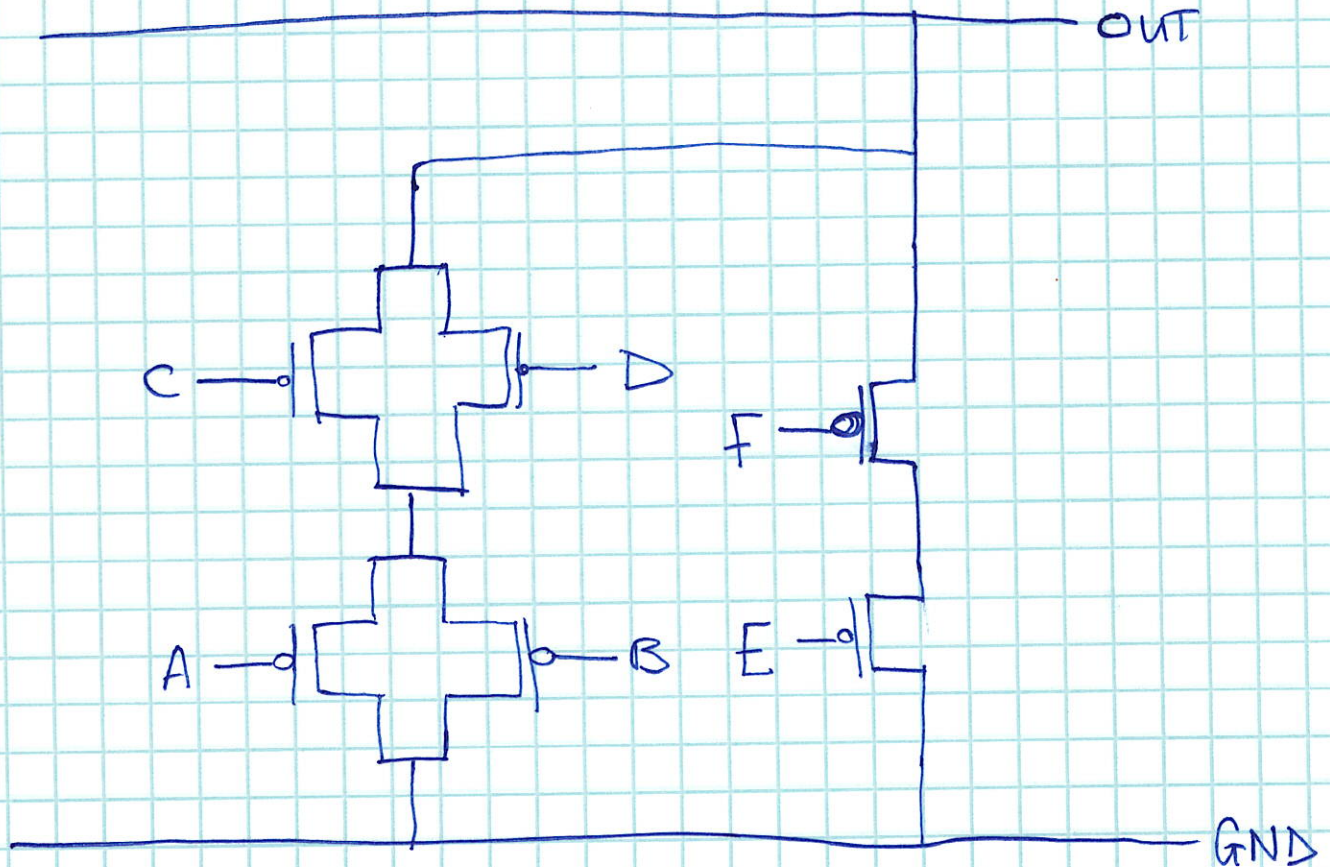
* Notes Pg 8, Q1E
For Notes taken whilst
Rewatching lectures

Reasonable & acceptable levels of failure allow for optimization at different levels of abstraction in design. Area ^{requirements} are unlikely to be relaxed over much but perhaps frequency could be reduced to reduce P_{dyn} and therefore Power density.

different bonding techniques, in the actual manufacture of the chip, could help mitigate the increased costs from increased design complexity.

$$F = (A \text{ OR } B) \text{ AND } (C \text{ OR } D) \text{ OR } E \text{ AND } F$$

$$\overline{F} = (A \text{ OR } B)(C \text{ OR } D) \text{ OR } E \text{ AND } F$$



Notes Page 2,
Q 2a

Question 2: Static CMOS Logic Implementation and Sizing [20 points]

Implement the following Boolean function as a single static complementary CMOS gate:

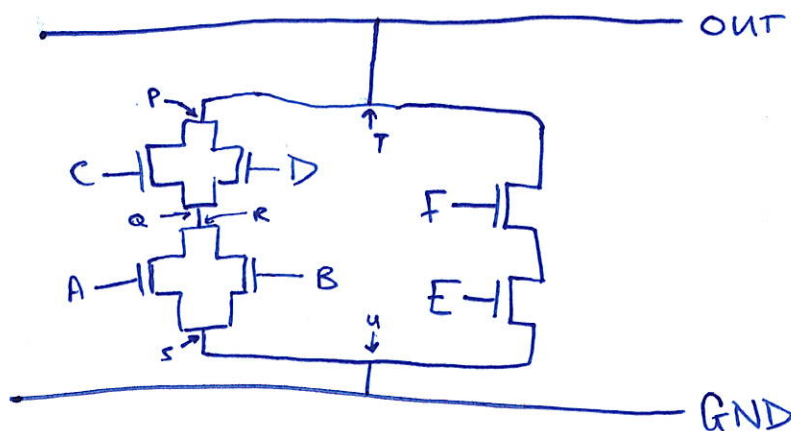
$$F = \overline{(A + B)(C + D) + EF}.$$

Do not decompose the function into multiple NAND/NOR/inverter gates. You may assume true and complemented input signals are not both available unless you explicitly add an inverter. For this problem, implement the function directly using only inputs A, B, C, D, E, F .

Assume:

- Unit width NMOS ON resistance is R_n .
- Unit width PMOS ON resistance is $2R_n$.
- Unit width NMOS input capacitance is C_g .
- Unit width PMOS input capacitance is $2C_g$.
- Transistor widths must be integer multiples of unit width.
- Ignore diffusion capacitance in the numerical parts.

- (a) **Pull-down network.** Draw the NMOS pull-down network that causes $F = 0$. Clearly label all transistor gates and internal nodes. [4 points]



$$X = \overline{F}$$

$$\text{When } F=0 \text{ } x=1$$

$$X = (A+B)(C+D) + EF$$

~~$$F=0 \text{ OR } E \neq 0.$$~~
~~$$C=0 \text{ AND } D$$~~

FOR output to = 0 :

$$F=1 \text{ and } E=1$$

$$C \text{ OR } D = 1$$

$$A \text{ OR } B = 1$$

T: Branched connection to C OR D & A OR B, and to E and F.

P: Branched connection to C OR D

R: Branched connection to A OR B

Q: ~~AND~~ C OR D

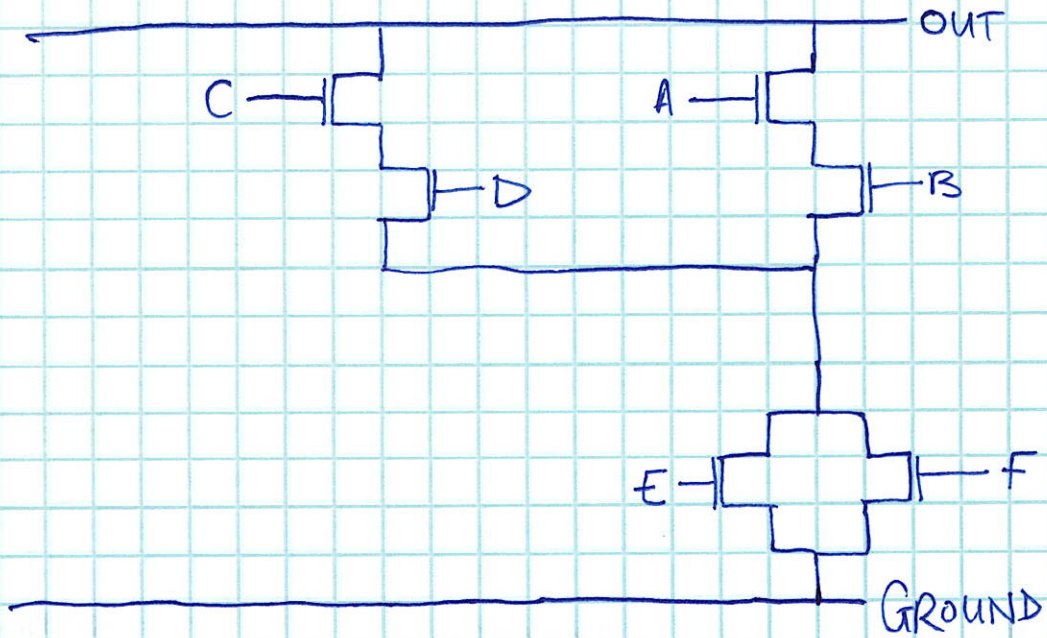
S: A OR B → GND

U: Branched connection to GND

More
Notes on
Note Page 2

NMOS STRONGER.
BOTTOM

if 0 then v.



$A = 0$
 $B = 0$
~~AND~~
OR

$C = 0$
 $D = 0$

AND
 $E = 0$
OR
 $F = 0$

Notes Page 3
Q2b